1

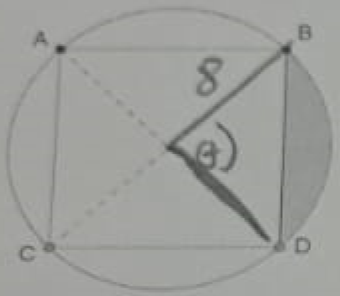
$$S(\theta) - C(\theta) = S(6\theta)$$

$$(180-\theta) - (90-\theta) = (180-6\theta)$$

$$-90 = -6\theta$$

$$\boxed{\theta = 15}$$

2. Hallar el área sombreada de la figura, donde ABCD es un cuadrado inscrito en una circunferencia de radio 8cm.



$\theta = \frac{\pi}{2}$

$R = 8$

$$A_S = A_{\triangleleft} - A_{\triangle}$$

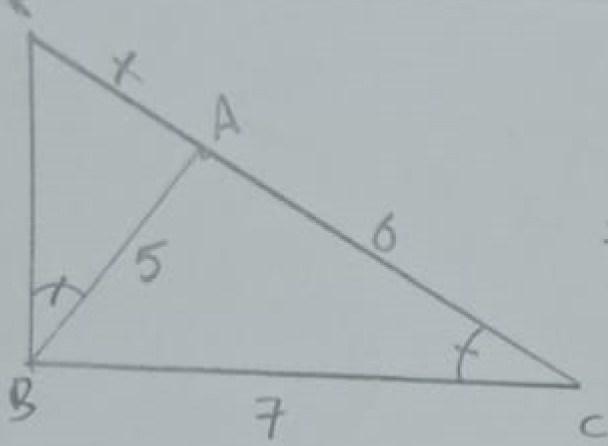
$$= \frac{1}{2}\left(\frac{\pi}{2}\right)8^2 - \frac{1}{2}8^2$$

$$= 32\left(\frac{\pi}{2} - 1\right)$$

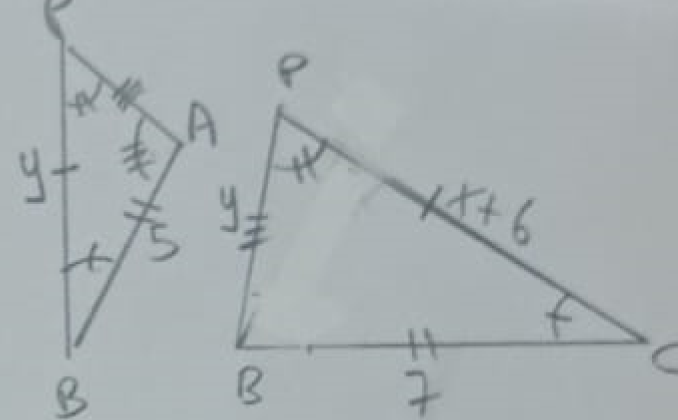
$$= 16(\pi - 2)$$

A) 16(π-1) B) 16(π-2) C) 16(π-3) D) 16(π-4) E) ninguno

(3)



$\Rightarrow$



$\Delta APB \sim \Delta PBC$

$$\frac{PB}{x+6} = \frac{5}{7} = \frac{x}{PB}$$

$$\begin{cases} \dfrac{PB}{x+6} = \dfrac{x}{PB} \\ \dfrac{x}{PB} = \dfrac{5}{7} \end{cases} \Rightarrow \begin{cases} PB^2 = x^2+6x \\ PB = \dfrac{7x}{5} \end{cases} \Rightarrow \left(\frac{7x}{5}\right)^2 = x^2+6x$$

$$24x^2 - 150x = 0$$

$$x(24x-150)$$

$$x=0,\ x=\frac{25}{4} \Rightarrow \boxed{x=\frac{25}{4}}$$

4

Polígono 1

n

$$D_1 = \frac{n(n-3)}{2}$$

$i = ?$

Polígono 2

$m = n-1$

$$D_2 = \frac{(n-1)((n-1)-3)}{2} = \frac{(n-1)(n-4)}{2}$$

$$D_1 = D_2 + 7$$

$$\frac{n(n-3)}{2} = \frac{(n-1)(n-4)}{2} + 7$$

$$n^2 - 3n = n^2 - 5n + 4 + 14$$

$$2n = 18$$

$$\underline{n = 9}$$

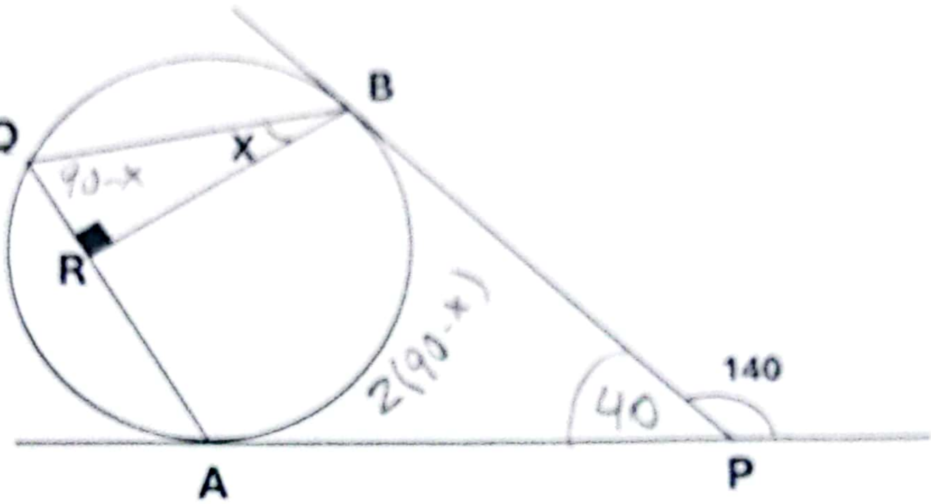
Luego.

$$i = \frac{(n-2)180^\circ}{n}$$

$$i = \frac{(9-2)180^\circ}{9}$$

$$\underline{i = 140^\circ}$$

5. Hallar el valor de "x" si A y B son puntos de tangencia



a) 20 b) 70 c) 40 d) 35 e) Ninguno

Ángulo externo

$$\alpha = \frac{\overset{\frown}{AQB} - \overset{\frown}{AB}}{2}$$

$$40^\circ = \frac{(360^\circ - 2(90-x)) - 2(90^\circ - x)}{2}$$

$$80^\circ = 360^\circ - 180^\circ + 2x - 180^\circ + 2x$$

$$80^\circ = 4x$$

$$\boxed{20^\circ = x}$$